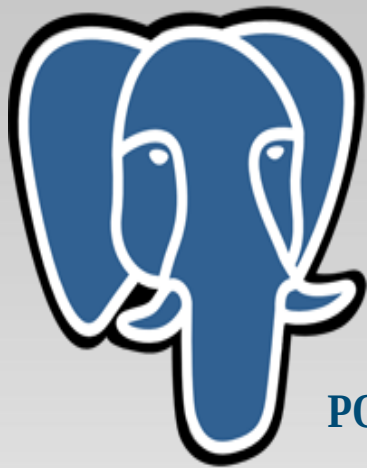




PostgreSQL

POSTGRES DATABASE ADMINISTRATION



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[SENIOR DATABASE ADMINISTRATOR|ORACLE| POSTGRESQL | MSSQL|OCI|AWS]

Chapter 4: PostgreSQL Management of Database Server

About the Author:

I am **Stephen Njoroge**, a Senior Database Engineer with over 8+ years of expertise in designing, securing and optimizing high-performance database systems across the banking and fintech sectors. My core technical proficiency spans major databases: Oracle, PostgreSQL, MS SQL Server and MySQL-along with advanced scripting in SQL and Python for automation and data pipeline development. I architect and manage cloud and virtualization solutions on OCI, AWS, and Proxmox for onprem systems and implement comprehensive DevOps practices using monitoring tools like Prometheus and Grafana and IT service management via iTop. Further, I deliver advanced data analytics and visualization using Tableau, Power BI and Oracle BI. As a Certified Data Protection Officer, I integrate robust data protection, compliance and security measures including vulnerability assessment and role-based access control into every layer of the data infrastructure.

PostgreSQL Database Server Management

Creating and Managing Databases

- Using OS Command

```
# Connect to PostgreSQL
psql -U postgres
```

```
postgres@998-986-02:/home/steve$ psql -U postgres
psql (18.1 (Ubuntu 18.1-1.pgdg24.04+2))
Type "help" for help.

postgres=# |
```

```
# Create database from OS command Line
--First create the user/role
```

```
createuser -U postgres analytics_user
```

```
postgres@998-986-02:/home/steve$ createuser -U postgres analytics_user
postgres@998-986-02:/home/steve$ psql
psql (18.1 (Ubuntu 18.1-1.pgdg24.04+2))
Type "help" for help.
```

```
postgres=# \du

          List of roles

```

Role name	Attributes
analytics_db	
analytics_user	
archive_user	
dev_user	
postgres	Superuser, Create role, Create DB, Replication, Bypass RLS
prod_user	
reporting_user	
training_user	

```
postgres=#
```

```
--create the database with that owner
```

```
createdb -U postgres -O analytics_user analytics_db
```

```

postgres@998-986-02:/home/steve$ createdb -U postgres -O analytics_user analytics_db
postgres@998-986-02:/home/steve$ psql
psql (18.1 (Ubuntu 18.1-1.pgdg24.04+2))
Type "help" for help.

postgres=# \l
postgres=# |

```

Verify database creation

```
psql -U postgres -c "\l"
```

```

postgres@998-986-02:/home/steve$ psql -U postgres -c "\l"

```

Name	Owner	Encoding	Locale Provider	Collate	Ctype	Locale	ICU Rules	Access privileges
analytics_db	analytics_user	UTF8	libc	C.UTF-8	C.UTF-8			
app_template	postgres	UTF8	libc	C.UTF-8	C.UTF-8			
my_custom_template	postgres	UTF8	libc	C.UTF-8	C.UTF-8			
postgres	postgres	UTF8	libc	C.UTF-8	C.UTF-8			
template0	postgres	UTF8	libc	C.UTF-8	C.UTF-8			=c/postgres +
								postgres=CTc/postgres
template1	postgres	UTF8	libc	C.UTF-8	C.UTF-8			=c/postgres +
								postgres=CTc/postgres

```

(6 rows)

postgres@998-986-02:/home/steve$

```

■ Creating Database from psql

-- Connect to PostgreSQL

```
psql -U postgres
```

-- Create a simple database

```
CREATE DATABASE training_db;
```

```

postgres@998-986-02:/home/steve$ psql -U postgres
psql (18.1 (Ubuntu 18.1-1.pgdg24.04+2))
Type "help" for help.

postgres=# CREATE DATABASE training_db;
CREATE DATABASE
postgres=#

```

-- Verify creation

```
\l
```

-- Connect to the new database

```
\c training_db
```

```
postgres=# \l
```

Name	Owner	Encoding	List of databases				Locale	ICU Rules	Access privileges
			Locale	Provider	Collate	Ctype			
analytics_db	analytics_user	UTF8	libc		C.UTF-8	C.UTF-8			
app_template	postgres	UTF8	libc		C.UTF-8	C.UTF-8			
my_custom_template	postgres	UTF8	libc		C.UTF-8	C.UTF-8			
postgres	postgres	UTF8	libc		C.UTF-8	C.UTF-8			
template0	postgres	UTF8	libc		C.UTF-8	C.UTF-8			=c/postgres + postgres=CtC/postgres
template1	postgres	UTF8	libc		C.UTF-8	C.UTF-8			=c/postgres + postgres=CtC/postgres
training_db	postgres	UTF8	libc		C.UTF-8	C.UTF-8			

```
(7 rows)
```

```
postgres=# \c training_db
You are now connected to database "training_db" as user "postgres".
training_db=#
```

■ Create database with specific owner

```
CREATE DATABASE sales_db OWNER sales_user;
```

```
training_db=# CREATE DATABASE sales_db OWNER sales_user;
CREATE DATABASE
training_db=# \l
```

Name	Owner	Encoding	List of databases				Locale	ICU Rules	Access privileges
			Locale	Provider	Collate	Ctype			
analytics_db	analytics_user	UTF8	libc		C.UTF-8	C.UTF-8			
app_template	postgres	UTF8	libc		C.UTF-8	C.UTF-8			
my_custom_template	postgres	UTF8	libc		C.UTF-8	C.UTF-8			
postgres	postgres	UTF8	libc		C.UTF-8	C.UTF-8			
sales_db	sales_user	UTF8	libc		C.UTF-8	C.UTF-8			
template0	postgres	UTF8	libc		C.UTF-8	C.UTF-8			=c/postgres + postgres=CtC/postgres
template1	postgres	UTF8	libc		C.UTF-8	C.UTF-8			=c/postgres + postgres=CtC/postgres
training_db	postgres	UTF8	libc		C.UTF-8	C.UTF-8			

```
(8 rows)
```

```
training_db=# |
```

```
-- Create database with encoding
```

```
CREATE DATABASE europe_db
ENCODING 'UTF8';
```

```
training_db=# \l
```

Name	Owner	Encoding	List of databases				Locale	ICU Rules	Access privileges
			Locale	Provider	Collate	Ctype			
analytics_db	analytics_user	UTF8	libc		C.UTF-8	C.UTF-8			
app_template	postgres	UTF8	libc		C.UTF-8	C.UTF-8			
europe_db	postgres	UTF8	libc		C.UTF-8	C.UTF-8			
my_custom_template	postgres	UTF8	libc		C.UTF-8	C.UTF-8			
postgres	postgres	UTF8	libc		C.UTF-8	C.UTF-8			
sales_db	sales_user	UTF8	libc		C.UTF-8	C.UTF-8			
template0	postgres	UTF8	libc		C.UTF-8	C.UTF-8			=c/postgres + postgres=CtC/postgres
template1	postgres	UTF8	libc		C.UTF-8	C.UTF-8			=c/postgres + postgres=CtC/postgres
training_db	postgres	UTF8	libc		C.UTF-8	C.UTF-8			

```
(9 rows)
```

```
training_db=# |
```

```
-- Create database with connection limit
```

```
CREATE DATABASE limited_db
CONNECTION LIMIT 10;
```

```
postgres=# CREATE DATABASE limited_db
CONNECTION LIMIT 10;
CREATE DATABASE
postgres=# \l
```

List of databases									
Name	Owner	Encoding	Locale Provider	Collate	Ctype	Locale	ICU Rules	Access privileges	
analytics_db	analytics_user	UTF8	libc	C.UTF-8	C.UTF-8				
app_template	postgres	UTF8	libc	C.UTF-8	C.UTF-8				
europe_db	postgres	UTF8	libc	C.UTF-8	C.UTF-8				
limited_db	postgres	UTF8	libc	C.UTF-8	C.UTF-8				
my_custom_template	postgres	UTF8	libc	C.UTF-8	C.UTF-8				
postgres	postgres	UTF8	libc	C.UTF-8	C.UTF-8				
sales_db	sales_user	UTF8	libc	C.UTF-8	C.UTF-8				
template0	postgres	UTF8	libc	C.UTF-8	C.UTF-8			=c/postgres	+
template1	postgres	UTF8	libc	C.UTF-8	C.UTF-8			postgres=CTc/postgres	+
training_db	postgres	UTF8	libc	C.UTF-8	C.UTF-8			=c/postgres	+
(10 rows)								postgres=CTc/postgres	

```
postgres=#
```

▪ Setting Database Parameters

-- Alter database parameters after creation

```
ALTER DATABASE analytics_db SET search_path TO analytics, public;
ALTER DATABASE analytics_db SET work_mem = '64MB';
ALTER DATABASE analytics_db SET maintenance_work_mem = '256MB';
ALTER DATABASE analytics_db SET effective_cache_size = '4GB';
```

```
postgres=# ALTER DATABASE analytics_db SET search_path TO analytics, public;
ALTER DATABASE analytics_db SET work_mem = '64MB';
ALTER DATABASE analytics_db SET maintenance_work_mem = '256MB';
ALTER DATABASE analytics_db SET effective_cache_size = '4GB';
ALTER DATABASE
ALTER DATABASE
ALTER DATABASE
ALTER DATABASE
postgres=#
```

-- View database parameters

```
SELECT name, setting FROM pg_settings
WHERE name IN ('search_path', 'work_mem', 'maintenance_work_mem')
UNION ALL
SELECT 'effective_cache_size', setting
FROM pg_settings
WHERE name = 'effective_cache_size';
```

```

postgres=# SELECT name, setting FROM pg_settings
WHERE name IN ('search_path', 'work_mem', 'maintenance_work_mem')
UNION ALL
SELECT 'effective_cache_size', setting
FROM pg_settings
WHERE name = 'effective_cache_size';
      name      |      setting
-----+-----
maintenance_work_mem | 65536
search_path      | "$user", public
work_mem         | 4096
effective_cache_size | 524288
(4 rows)

postgres=#

```

Schema Management

■ Creating and Using Schemas

```

-- Create a new schema
CREATE SCHEMA hr;

```

```

postgres=# CREATE SCHEMA hr;
CREATE SCHEMA
postgres=# |

```

```

-- Create schema with specific owner
CREATE SCHEMA finance AUTHORIZATION finance_user;

```

```

postgres=# CREATE SCHEMA finance AUTHORIZATION finance_user;
CREATE SCHEMA
postgres=#

```

```

-- Create table in specific schema
CREATE TABLE hr.employees (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100),
    department VARCHAR(50),
    salary DECIMAL(10,2)
);

```

```

postgres=# CREATE TABLE hr.employees (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100),
    department VARCHAR(50),
    salary DECIMAL(10,2)
);
CREATE TABLE
postgres=#

```

```
-- List all schemas
\dn
```

```
postgres=# \dn
      List of schemas
  Name | Owner
-----+-----
 finance | finance_user
  hr      | postgres
 public   | pg_database_owner
tenant_a | postgres
tenant_b | postgres
(5 rows)

postgres=#
```

Accessing Schema Objects

```
-- Access table with schema prefix
SELECT * FROM hr.employees ORDER BY id;
```

```
postgres=# SELECT * FROM hr.employees ORDER BY id;
 id | name | department | salary
----+-----+-----+-----
  1 | John Smith | Engineering | 85000.00
  2 | Sarah Johnson | Marketing | 72000.00
  3 | Michael Chen | Engineering | 92000.00
  4 | Emily Davis | Human Resources | 65000.00
  5 | Robert Williams | Sales | 78000.00
  6 | Jennifer Lee | Engineering | 88000.00
  7 | David Brown | Finance | 95000.00
  8 | Lisa Wilson | Marketing | 68000.00
  9 | James Taylor | Operations | 72000.00
 10 | Maria Garcia | Engineering | 102000.00
 11 | Thomas Miller | Sales | 81000.00
 12 | Patricia Jones | Human Resources | 67000.00
 13 | Christopher Moore | Engineering | 89000.00
```

```
-- Set search path for easier access
SET search_path TO hr, public;
```

```
-- View current search path
SHOW search_path;
```

```

postgres=# SET search_path TO hr, public;
SET
postgres=# SHOW search_path;
search_path
-----
hr, public
(1 row)

postgres=#

```

User Management

▪ Creating Users with Attributes

```

-- Create user with password
CREATE USER developer WITH PASSWORD 'SecurePassdev123!';

```

```

postgres=# CREATE USER developer WITH PASSWORD 'SecurePassdev123!';
CREATE ROLE
postgres=# \du

```

Role name	Attributes
analytics_db	
analytics_user	
archive_user	
dev_user	
developer	
finance_user	
postgres	Superuser, Create role, Create DB, Replication, Bypass RLS
prod_user	
reporting_user	
sales_user	
training_user	

```

postgres=# |

```

```

-- Create user with additional attributes
CREATE USER analyst_user WITH
  PASSWORD 'AnalystPass456!'
  VALID UNTIL '2026-12-31'
  CONNECTION LIMIT 10;

```

```
postgres=# CREATE USER analyst_user WITH
  PASSWORD 'AnalystPass456!'
  VALID UNTIL '2026-12-31'
  CONNECTION LIMIT 10;
CREATE ROLE
postgres=# \du
                                List of roles
  Role name | Attributes
-----+-----
analyst     | 10 connections +
            | Password valid until 2024-12-31 00:00:00+03
analyst_user | 10 connections +
            | Password valid until 2026-12-31 00:00:00+03
analytics_db
analytics_user
archive_user
dev_user
developer
finance_user
postgres    | Superuser, Create role, Create DB, Replication, Bypass RLS
prod_user
reporting_user
sales_user
training_user

postgres=#
```

```
-- Create superuser
CREATE USER admin WITH
  PASSWORD 'AdminPass789!'
  SUPERUSER
  CREATEDB
  CREATEROLE;
```

```
postgres=# CREATE USER admin WITH
  PASSWORD 'AdminPass789!'
  SUPERUSER
  CREATEDB
  CREATEROLE;
CREATE ROLE
postgres=# \du
                                List of roles
  Role name | Attributes
-----+-----
admin      | Superuser, Create role, Create DB
analyst    | 10 connections +
            | Password valid until 2024-12-31 00:00:00+03
analyst_user | 10 connections +
            | Password valid until 2026-12-31 00:00:00+03
analytics_db
analytics_user
archive_user
dev_user
developer
finance_user
postgres    | Superuser, Create role, Create DB, Replication, Bypass RLS
prod_user
reporting_user
sales_user
training_user

postgres=# |
```

▪ User Renaming and Modification

```
-- Rename user
ALTER USER developer RENAME TO senior_dev;
```

```
postgres=# ALTER USER developer RENAME TO senior_dev;
ALTER ROLE
postgres=# \du

```

Role name	List of roles	Attributes
admin	Superuser, Create role, Create DB	
analyst	10 connections	+
analyst_user	Password valid until 2024-12-31 00:00:00+03	
	10 connections	+
	Password valid until 2026-12-31 00:00:00+03	
analytics_db		
analytics_user		
archive_user		
dev_user		
finance_user		
postgres	Superuser, Create role, Create DB, Replication, Bypass RLS	
prod_user		
reporting_user		
sales_user		
senior_dev		
training_user		

```
postgres=# |
```

```
-- Change password
ALTER USER senior_dev WITH PASSWORD 'NewSecurePass321!';
```

```
postgres=# ALTER USER senior_dev WITH PASSWORD 'NewSecurePass321!';
ALTER ROLE
postgres=# |
```

```
-- Modify user attributes
ALTER USER analyst
  VALID UNTIL '2025-06-30'
  CONNECTION LIMIT 20;
```

```
postgres=# ALTER USER analyst
VALID UNTIL '2025-06-30'
CONNECTION LIMIT 20;
ALTER ROLE
postgres=# \du
```

List of roles	
Role name	Attributes
admin	Superuser, Create role, Create DB
analyst	20 connections +
	Password valid until 2025-06-30 00:00:00+03
analyst_user	10 connections +
	Password valid until 2026-12-31 00:00:00+03
analytics_db	
analytics_user	
archive_user	
dev_user	
finance_user	
postgres	Superuser, Create role, Create DB, Replication, Bypass RLS
prod_user	
reporting_user	
sales_user	
senior_dev	
training_user	

```
postgres=# |
```

```
-- Grant privileges
GRANT ALL PRIVILEGES ON DATABASE limited_db TO senior_dev;
GRANT USAGE ON SCHEMA hr TO analyst;
GRANT SELECT ON hr.employees TO analyst;
```

```
postgres=# GRANT ALL PRIVILEGES ON DATABASE limited_db TO senior_dev;
GRANT USAGE ON SCHEMA hr TO analyst;
GRANT SELECT ON hr.employees TO analyst;
GRANT
GRANT
GRANT
postgres=# |
```

Schema Ownership

```
-- Transfer schema ownership
ALTER SCHEMA hr OWNER TO hr_manager;

-- Create user for HR management
CREATE USER hr_manager WITH PASSWORD 'HrMgrPass123';

-- Grant privileges on schema objects
GRANT ALL PRIVILEGES ON SCHEMA hr TO hr_manager;
GRANT ALL PRIVILEGES ON ALL TABLES IN SCHEMA hr TO hr_manager;

-- Verify ownership
\dn+
```

```

postgres=# create user hr_manager;
CREATE ROLE
postgres=# ALTER SCHEMA hr OWNER TO hr_manager;
ALTER SCHEMA
postgres=# CREATE USER hr_manager WITH PASSWORD 'HrMgrPass123';
ERROR:  role "hr_manager" already exists
postgres=# GRANT ALL PRIVILEGES ON SCHEMA hr TO hr_manager;
GRANT ALL PRIVILEGES ON ALL TABLES IN SCHEMA hr TO hr_manager;
GRANT
GRANT
postgres=# \dn+

```

List of schemas			
Name	Owner	Access privileges	Description
finance	finance_user		
hr	hr_manager	hr_manager=UC/hr_manager analyst=U/hr_manager	
public	pg_database_owner	pg_database_owner=UC/pg_database_owner+ =U/pg_database_owner	standard public schema
tenant_a	postgres		
tenant_b	postgres		
(5 rows)			

```
postgres=#
```

Local and Remote Connectivity

Local Connectivity

```

# Local connection using peer authentication (default)
psql -U postgres -d analytics_db

```

```

postgres@998-986-02:/home/steve$ psql -U postgres -d analytics_db
psql (18.1 (Ubuntu 18.1-1.pgdg24.04+2))
Type "help" for help.

analytics_db=#

```

```

# Local connection specifying host
psql -h localhost -U postgres -d analytics_db

```

```

postgres@998-986-02:/home/steve$ psql -h localhost -U postgres -d analytics_db
Password for user postgres:
psql (18.1 (Ubuntu 18.1-1.pgdg24.04+2))
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, compression: off, ALPN: postgresql)
Type "help" for help.

analytics_db=#

```

Understanding pg_hba.conf

```

# Locate pg_hba.conf file
sudo find / -name "pg_hba.conf" 2>/dev/null
# Typical locations: /etc/postgresql/XX/main/pg_hba.conf or /var/lib/postgresql/data/pg_hba.conf

```

```

root@998-986-02:/home/steve# sudo find / -name "pg_hba.conf" 2>/dev/null
/etc/postgresql/18/main/pg_hba.conf
/etc/postgresql/18/prod/pg_hba.conf
/etc/postgresql/18/dev/pg_hba.conf
/var/lib/postgresql/mydata/pg_hba.conf
/var/lib/postgresql/18/data/pg_hba.conf

```

Backup current configuration

```

sudo cp /etc/postgresql/18/main/pg_hba.conf /etc/postgresql/18/main/pg_hba.conf.backup

```

▪ pg_hba.conf Configuration

```

vi /etc/postgresql/18/main/pg_hba.conf

```

```

#
# Database administrative login by Unix domain socket
local  all             postgres                                peer

# TYPE  DATABASE        USER            ADDRESS              METHOD

# "local" is for Unix domain socket connections only
local  all             all              peer
# IPv4 local connections:
host   all             all              127.0.0.1/32         scram-sha-256
# IPv6 local connections:
host   all             all              ::1/128              scram-sha-256
# Allow replication connections from localhost, by a user with the
# replication privilege.
local  replication     all              peer
host   replication     all              127.0.0.1/32         scram-sha-256
host   replication     all              ::1/128              scram-sha-256

```

▪ Authentication Methods

```

SELECT pg_reload_conf();

```

```

postgres@998-986-02:/home/steve$ psql
psql (18.1 (Ubuntu 18.1-1.pgdg24.04+2))
Type "help" for help.

```

```

postgres=# SELECT pg_reload_conf();
 pg_reload_conf

```

```

-----

```

```

 t
(1 row)

```

```

postgres=# |

```

Predefined Roles

-- View predefined roles

```

SELECT rolname, rolsuper, rolcreatorole, rolcreatedb
FROM pg_roles
WHERE rolname LIKE 'pg_%';

```

```
postgres=# SELECT rolname, rolsuper, rolcreatorole, rolcreatedb
FROM pg_roles
WHERE rolname LIKE 'pg_%';
```

rolname	rolsuper	rolcreatorole	rolcreatedb
pg_database_owner	f	f	f
pg_read_all_data	f	f	f
pg_write_all_data	f	f	f
pg_monitor	f	f	f
pg_read_all_settings	f	f	f
pg_read_all_stats	f	f	f
pg_stat_scan_tables	f	f	f
pg_read_server_files	f	f	f
pg_write_server_files	f	f	f
pg_execute_server_program	f	f	f
pg_signal_backend	f	f	f
pg_checkpoint	f	f	f
pg_maintain	f	f	f
pg_use_reserved_connections	f	f	f
pg_create_subscription	f	f	f
pg_signal_autovacuum_worker	f	f	f

```
(16 rows)

postgres=#
```

Extensions in PostgreSQL

```
-- List available extensions
SELECT * FROM pg_available_extensions;
```

name	default_version	installed_version	comment
-----+-----+-----+-----			
address_standardizer_data_us-3	3.6.1		Address Standardizer US dataset example
pg_trgm	1.6		text similarity measurement and index searching based on trig
rams			
pg_buffercache	1.6		examine the shared buffer cache
autoinc	1.0		functions for autoincrementing fields
btree_gin	1.3		support for indexing common datatypes in GIN
citext	1.8		data type for case-insensitive character strings
fuzzystrmatch	1.2		determine similarities and distance between strings
pgcrypto	1.4		cryptographic functions
oracle_fdw	1.2		foreign data wrapper for Oracle access
postgis_tiger_geocoder-3	3.6.1		PostGIS tiger geocoder and reverse geocoder
dblink	1.2	1.2	connect to other PostgreSQL databases from within a database
sslnfo	1.2		information about SSL certificates
pgstattuple	1.5		show tuple-level statistics
tcn	1.0		Triggered change notifications
refint	1.0		functions for implementing referential integrity (obsolete)
address_standardizer_data_us	3.6.1		Address Standardizer US dataset example
postgis_topology-3	3.6.1		PostGIS topology spatial types and functions
tsm_system_time	1.0		TABLESAMPLE method which accepts time in milliseconds as a li
mit			
address_standardizer	3.6.1		Used to parse an address into constituent elements. Generally
used to support geocoding address normalization step.			
amcheck	1.5		functions for verifying relation integrity
postgres_fdw	1.2		foreign-data wrapper for remote PostgreSQL servers
cube	1.5		data type for multidimensional cubes
seg	1.4		data type for representing line segments or floating-point in
tervals			

```
-- List installed extensions
\d+
```

```
postgres=# \dx
```

List of installed extensions				
Name	Version	Default version	Schema	Description
dblink	1.2	1.2	public	connect to other PostgreSQL databases from within a database
plpgsql	1.0	1.0	pg_catalog	PL/pgSQL procedural language

```
(2 rows)
```

```
postgres=#
```

```
-- Install popular extensions
```

```
CREATE EXTENSION IF NOT EXISTS "uuid-ossf";
```

```
CREATE EXTENSION IF NOT EXISTS "pgcrypto";
```

```
CREATE EXTENSION IF NOT EXISTS "postgis";
```

```
postgres=# CREATE EXTENSION IF NOT EXISTS "uuid-ossf";
```

```
CREATE EXTENSION IF NOT EXISTS "pgcrypto";
```

```
CREATE EXTENSION IF NOT EXISTS "postgis";
```

```
CREATE EXTENSION
```

```
CREATE EXTENSION
```

```
CREATE EXTENSION
```

```
postgres=#
```

```
-- Use UUID extension
```

```
SELECT uuid_generate_v4();
```

```
postgres=# SELECT uuid_generate_v4();
```

```
uuid_generate_v4
```

```
930cf84a-8faa-4c94-8eee-108703737f5a
```

```
(1 row)
```

```
postgres=# |
```

```
-- Use pgcrypto for password hashing
```

```
SELECT crypt('passberry321!', gen_salt('md5'));
```

```
postgres=# SELECT crypt('passberry321!', gen_salt('md5'));
```

```
crypt
```

```
$1$c2HHVewu$hVJb3YVsn0m78CPhtquqx0
```

```
(1 row)
```

```
postgres=#
```

Cleanup Operations

▪ Dropping Users

```
-- Reassign objects to another user
REASSIGN OWNED BY analyst TO senior_dev;
```

```
postgres=# REASSIGN OWNED BY analyst TO senior_dev;
REASSIGN OWNED
postgres=#
```

```
-- Drop owned objects
DROP OWNED BY analyst;
```

```
postgres=# DROP OWNED BY analyst;
DROP OWNED
postgres=#
```

```
-- Drop user
DROP USER IF EXISTS analyst;
```

```
postgres=# DROP USER IF EXISTS analyst;
DROP ROLE
postgres=#
```

▪ Dropping Schemas

```
-- Check schema contents
\dn+
```

```
postgres=# \dn+
```

Name	Owner	List of schemas Access privileges	Description
finance	finance_user		
hr	hr_manager	hr_manager=UC/hr_manager	
public	pg_database_owner	pg_database_owner=UC/pg_database_owner+ =U/pg_database_owner	standard public schema
tenant_a	postgres		
tenant_b	postgres		
(5 rows)			

```
postgres=#
```

```
-- Drop empty schema
DROP SCHEMA IF EXISTS sales;
```

```
postgres=# DROP SCHEMA IF EXISTS sales;
DROP SCHEMA
postgres=#
```

▪ Dropping Database Using OS Command

Drop database from command line

```
dropdb -U postgres sales_db
```

```
postgres@998-986-02:/home/steve$ dropdb -U postgres sales_db
```

```
postgres@998-986-02:/home/steve$ psql
```

```
psql (18.1 (Ubuntu 18.1-1.pgdg24.04+2))
```

```
Type "help" for help.
```

```
postgres=# \du
```

List of roles	
Role name	Attributes
admin	Superuser, Create role, Create DB
analyst_user	10 connections + Password valid until 2026-12-31 00:00:00+03
analytics_db	
analytics_user	
archive_user	
dev_user	
finance_user	
hr_manager	
postgres	Superuser, Create role, Create DB, Replication, Bypass RLS
prod_user	
reporting_user	
sales_user	
senior_dev	
training_user	

Drop database with verification

```
dropdb -U postgres --if-exists sales_db
```

Alternative: Drop from psql

```
psql -U postgres -c "DROP DATABASE IF EXISTS sales_db;"
```

```
postgres@998-986-02:/home/steve$ psql -U postgres -c "CREATE DATABASE sales_db;"
```

```
CREATE DATABASE
```

```
postgres@998-986-02:/home/steve$ psql -U postgres -c "DROP DATABASE IF EXISTS sales_db;"
```

```
DROP DATABASE
```

```
postgres@998-986-02:/home/steve$ |
```

Verify databases

```
psql -U postgres -c "\l"
```

List of databases								
Name	Owner	Encoding	Locale Provider	Collate	Ctype	Locale	ICU Rules	Access privileges
analytics_db	analytics_user	UTF8	libc	C.UTF-8	C.UTF-8			
app_template	postgres	UTF8	libc	C.UTF-8	C.UTF-8			
europe_db	postgres	UTF8	libc	C.UTF-8	C.UTF-8			
limited_db	postgres	UTF8	libc	C.UTF-8	C.UTF-8			=Tc/postgres + postgres=CTc/postgres + senior_dev=CTc/postgres
my_custom_template	postgres	UTF8	libc	C.UTF-8	C.UTF-8			
postgres	postgres	UTF8	libc	C.UTF-8	C.UTF-8			
template0	postgres	UTF8	libc	C.UTF-8	C.UTF-8			=c/postgres + postgres=CTc/postgres
template1	postgres	UTF8	libc	C.UTF-8	C.UTF-8			=c/postgres + postgres=CTc/postgres
training_db	postgres	UTF8	libc	C.UTF-8	C.UTF-8			
(9 rows)								

```
postgres@998-986-02:/home/steve$
```

Database Management Workflow

-- 1. Create database

```
CREATE DATABASE training_db WITH OWNER postgres;
```

```
postgres=# CREATE DATABASE training_db WITH OWNER postgres;
CREATE DATABASE
postgres=#
```

-- 2. Create schemas

```
\c training_db
```

```
CREATE SCHEMA inventory AUTHORIZATION postgres;
CREATE SCHEMA customers;
```

```
postgres=# CREATE DATABASE training_db WITH OWNER postgres;
CREATE DATABASE
postgres=# \c training_db
You are now connected to database "training_db" as user "postgres".
training_db=# CREATE SCHEMA inventory AUTHORIZATION postgres;
CREATE SCHEMA customers;
CREATE SCHEMA
CREATE SCHEMA
training_db=#
```

-- 3. Create users

```
CREATE USER inventory_mgr WITH PASSWORD 'InvPass123';
CREATE USER customer_svc WITH PASSWORD 'CustPass456';
```

```
training_db=# CREATE USER inventory_mgr WITH PASSWORD 'InvPass123';
CREATE USER customer_svc WITH PASSWORD 'CustPass456';
CREATE ROLE
CREATE ROLE
training_db=#
```

-- 4. Set ownership and privileges

```
ALTER SCHEMA inventory OWNER TO inventory_mgr;
GRANT USAGE ON SCHEMA customers TO customer_svc;
```

```
training_db=# ALTER SCHEMA inventory OWNER TO inventory_mgr;
GRANT USAGE ON SCHEMA customers TO customer_svc;
ALTER SCHEMA
GRANT
training_db=# |
```

-- 5. Create tables in schemas

```
CREATE TABLE inventory.products (  
    product_id SERIAL PRIMARY KEY,  
    name VARCHAR(100),  
    quantity INT  
);
```

```
CREATE TABLE customers.contacts (  
    contact_id SERIAL PRIMARY KEY,  
    email VARCHAR(255),  
    phone VARCHAR(20)  
);
```

```
training_db=# ALTER SCHEMA inventory OWNER TO inventory_mgr;  
GRANT USAGE ON SCHEMA customers TO customer_svc;  
ALTER SCHEMA  
GRANT
```

```
training_db=# CREATE TABLE inventory.products (  
    product_id SERIAL PRIMARY KEY,  
    name VARCHAR(100),  
    quantity INT  
);
```

```
CREATE TABLE customers.contacts (  
    contact_id SERIAL PRIMARY KEY,  
    email VARCHAR(255),  
    phone VARCHAR(20)  
);
```

```
CREATE TABLE  
CREATE TABLE  
training_db=#
```

-- 6. Install extension

```
CREATE EXTENSION "uuid-osspl";
```

```
training_db=# CREATE EXTENSION "uuid-osspl";  
CREATE EXTENSION  
training_db=#
```

-- 7. Grant predefined roles

```
GRANT pg_read_all_data TO customer_svc;
```

```
training_db=# GRANT pg_read_all_data TO customer_svc;  
GRANT ROLE  
training_db=# |
```